

USHIBUKA, Japan

WMO: 478380

Lat: 32.1978N

Lon: 130.0267E

Elev: 49

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 35.9	(c) 37.9	(d) 21.2	(e) 15.9	(f) 40.9	(g) 23.4	(h) 17.7	(i) 42.2	(j) 19.0	(k) 51.3	(l) 16.7	(m) 51.3	(n) 5.0	(o) 10	(p) 0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.8	(c) 91.5	(d) 78.6	(e) 89.9	(f) 78.5	(g) 88.3	(h) 78.0	(i) 80.8	(j) 87.3	(k) 80.0	(l) 86.4	(m) 79.3	(n) 85.4	(o) 9.0	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	151.2	84.4	78.3	147.2	83.8	77.6	143.9	83.3	44.6	87.7	43.7	86.7	43.0	85.6	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 17.9	(b) 15.1	(c) 13.2	(e) 32.2	(f) 95.8	(g) 2.0	(h) 2.0	(i) 30.7	(j) 97.2	(k) 29.6	(l) 98.4	(m) 28.5	(n) 99.5	(o) 27.0	(p) 101.0		
			DB													
			WB													
				28.4	81.9	1.9	0.9	27.0	82.6	26.0	83.1	24.9	83.6	23.5	84.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	47.3	49.0	54.0	61.5	68.6	74.1	81.2	83.7	78.9	70.7	61.0	51.4
	DBStd	13.18	5.14	5.71	5.23	4.64	3.37	3.30	3.17	2.69	3.89	4.55	5.67	5.73
	HDD50	269	114	80	21	0	0	0	0	0	0	0	1	53
	HDD65	2041	549	448	342	122	8	0	0	0	0	8	144	420
	CDD50	5822	30	52	144	345	577	723	968	1045	868	642	331	97
	CDD65	2118	0	0	0	16	119	273	503	580	418	185	24	0
	CDH74	17830	0	0	0	19	279	1276	5136	6824	3582	698	16	0
	CDH80	5621	0	0	0	1	14	148	1641	2740	997	80	0	0
(14) Wind	WSAvg	6.0	5.7	6.1	6.6	6.4	6.0	5.8	5.8	6.2	6.0	5.7	5.6	5.7
(15) Precipitation	PrecAvg	84.2	3.1	3.9	5.3	6.3	7.3	17.8	12.6	8.4	7.9	3.8	4.1	3.7
	PrecMax	135.2	6.2	8.0	13.0	15.6	14.5	40.1	36.5	26.4	17.7	11.2	8.5	6.7
	PrecMin	52.8	0.7	1.6	2.5	3.2	2.3	3.5	2.0	1.1	1.2	0.4	1.2	0.3
	PrecStd	18.3	1.2	1.8	2.1	2.9	3.6	7.8	8.1	5.3	5.2	2.8	2.0	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.1	66.7	70.7	76.4	81.7	85.8	92.7	95.1	91.1	85.5	76.1	67.6
		MCWB	60.0	60.9	60.8	64.3	68.0	75.7	78.9	78.2	77.2	73.9	67.1	60.6
	2%	DB	60.7	64.0	67.3	73.4	79.0	83.6	90.3	92.5	88.9	82.1	73.5	65.0
		MCWB	55.8	59.2	59.4	62.7	67.3	74.2	78.8	78.6	76.8	71.8	65.3	58.8
	5%	DB	57.9	61.3	65.0	71.3	77.0	81.6	88.8	90.8	87.0	79.9	71.4	62.6
		MCWB	51.8	56.3	58.3	62.1	66.3	73.4	78.5	78.7	75.9	69.9	64.1	56.0
	10%	DB	55.5	58.4	62.8	69.2	75.2	79.8	87.3	89.3	85.2	77.9	69.3	60.3
		MCWB	49.5	52.6	56.7	61.5	65.7	73.0	78.2	78.6	75.1	68.4	62.4	54.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.0	63.2	64.3	69.5	73.7	78.9	81.7	82.0	80.7	77.7	70.6	64.0
		MCDB	63.0	65.2	67.0	72.1	76.9	83.3	88.1	88.3	86.5	81.9	73.3	66.1
	2%	WB	56.9	60.4	61.8	67.1	71.2	77.5	80.6	81.0	79.1	74.6	67.9	60.0
		MCDB	59.9	63.2	65.2	70.1	74.4	81.0	86.8	87.7	85.0	79.9	71.0	63.6
	5%	WB	53.0	57.1	59.8	65.1	69.7	76.5	79.8	80.4	78.1	72.5	65.7	57.2
		MCDB	56.9	60.9	64.1	68.7	73.4	79.5	85.7	87.1	84.1	78.0	69.8	61.4
	10%	WB	49.9	53.6	57.5	63.1	68.2	75.3	79.1	79.7	77.0	70.2	63.5	54.7
		MCDB	54.8	57.7	62.1	67.4	72.8	78.1	84.7	86.3	83.1	76.0	68.7	59.7
(35) Mean Daily Temperature Range	MDBR		10.0	11.2	12.0	12.4	12.3	9.7	9.9	10.8	11.0	11.4	10.7	10.3
	5% DB	MCDBR	12.7	13.5	14.2	15.1	14.9	12.1	12.7	12.8	12.4	13.0	11.4	12.1
		MCWBR	10.4	11.7	10.0	8.5	7.4	5.4	4.9	4.7	5.2	6.5	7.5	9.3
	5% WB	MCDBR	12.0	12.3	12.1	10.8	10.4	7.6	10.6	10.8	10.5	10.3	9.4	10.7
		MCWBR	11.5	12.8	11.0	8.6	7.3	4.8	4.7	4.7	5.1	6.7	8.3	9.9
(40) Clear-Sky Solar Irradiance	taub		0.437	0.472	0.527	0.546	0.555	0.560	0.542	0.523	0.502	0.463	0.432	0.427
	taud		2.069	2.008	1.876	1.886	1.930	2.010	2.098	2.152	2.161	2.177	2.178	2.103
	Ebn at Noon		239	244	240	242	241	239	242	245	245	245	240	234
	Edh at Noon		43	50	61	63	61	56	51	48	46	42	38	39
(44) All-Sky Solar Radiation	RadAvg		699	932	1276	1525	1608	1339	1673	1722	1411	1171	842	658
	RadStd		72	92	84	137	180	220	212	138	147	120	82	59

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page